Smart LPG Gas Detection System

Enhancing Safety Through Technology — Real-Time Detection, Automatic Alerts & Intelligent Control

IOT SAFETY PROJECT | ARDUINO-BASED

Abstract

The Challenge

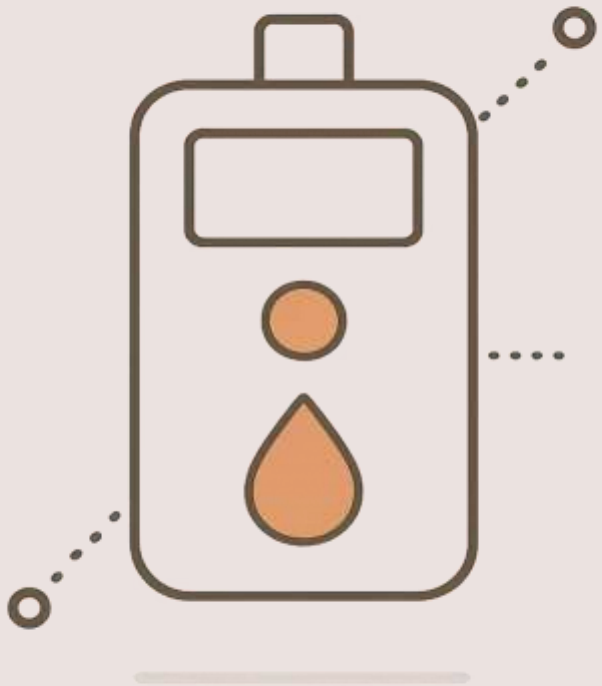
LPG gas leakage is a critical hazard in both domestic and industrial environments, posing risks of fire, explosion, and loss of life.

The Solution

This project proposes an intelligent, low-cost system for **real-time detection** and **automatic control** of LPG leaks using microcontroller-based sensing.

The Goal

Prevent hazardous situations by continuously monitoring gas levels and triggering immediate safety measures — alarms, notifications, and gas shut-off.



Objective

This system is designed with four core goals to deliver a complete, automated LPG safety solution.

Real-Time Detection

Design a low-cost, efficient system that continuously monitors LPG levels in real time.

Auto Shut-Off

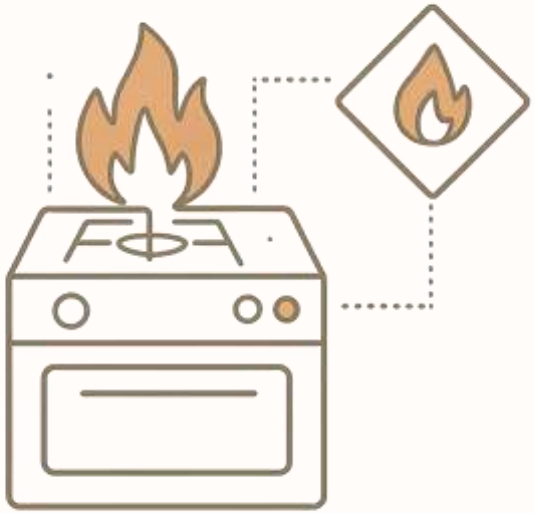
Automatically cut off the gas supply when leakage exceeds a predefined safety threshold.

Instant Alerts

Trigger an audible buzzer and visual LED indicator to ensure immediate user awareness.

Manual Control

Provide manual override options for added flexibility and user intervention when needed.



WHY IT MATTERS

Problem Statement

Severe Consequences

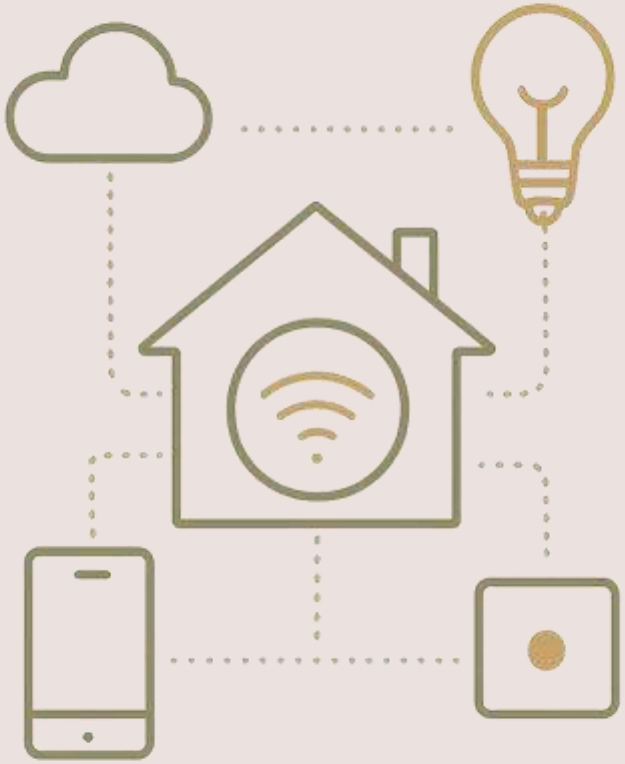
LPG leaks can cause fires, explosions, property damage, and fatal accidents — often within seconds.

Reactive, Not Proactive

Existing safety measures are often secondary or require manual intervention, leading to dangerously delayed responses.

The Gap

There is a clear need for an **automated, always-on system** that detects and mitigates LPG leaks before they escalate.



Introduction to Smart LPG Detection

LPG is one of the world's most widely used energy sources — but its volatile nature makes even small leaks a constant threat. Traditional detection methods are often slow, unreliable, or require constant human monitoring.

The Risk

LPG is odorless, colorless, and highly flammable — leaks go unnoticed until it's too late.

The Upgrade

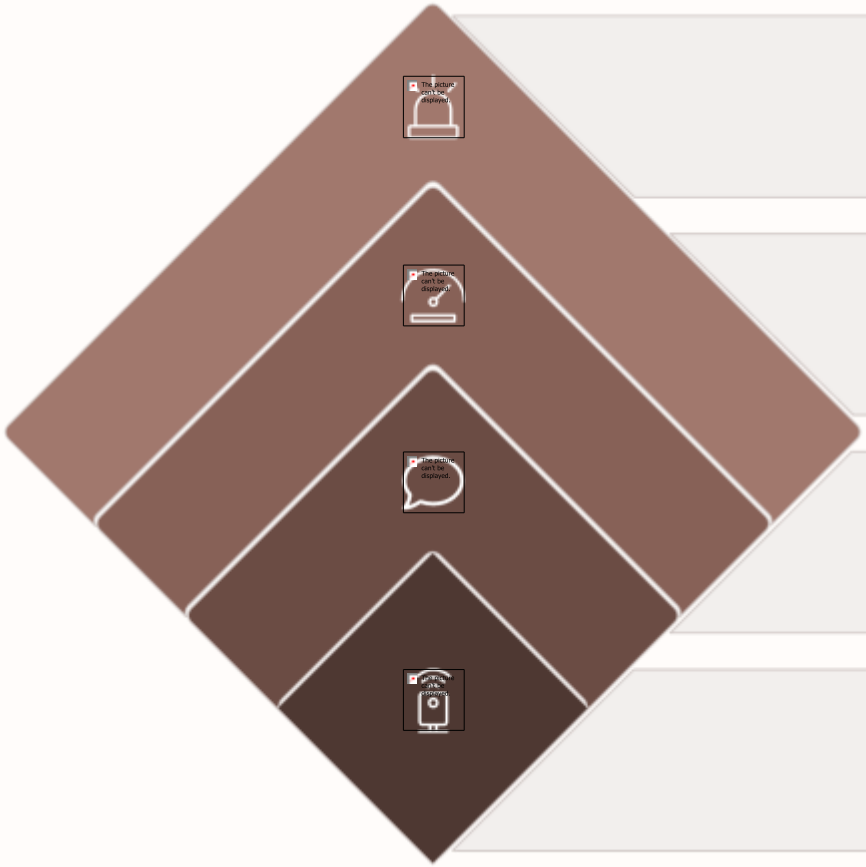
Smart systems use microcontrollers and gas sensors for continuous, automated safety management.

The Future

IoT integration enables remote monitoring, instant notifications, and intelligent response.

Arduino System Diagram

The system architecture centers on an **Arduino UNO** that processes sensor data and triggers safety responses when LPG levels exceed the threshold.



Outputs

Buzzer, LED, GSM optional

Threshold Check

Is gas > limit?

Arduino UNO

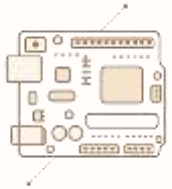
Processes sensor signal

MQ-2 Sensor

Detects LPG concentration

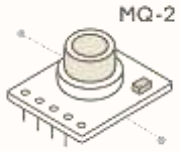
The MQ-2 sensor feeds real-time gas concentration data to the Arduino. When levels exceed the preset limit, the microcontroller simultaneously activates the buzzer, LED, and optional notification module for layered safety response.

Components



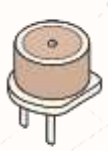
Arduino UNO

The brain of the system — reads sensor data and controls all outputs.



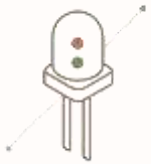
MQ-2 Gas Sensor

Detects LPG, propane, and flammable gases with high sensitivity.



Buzzer

Provides an audible warning alarm when gas is detected.

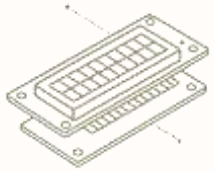


LED Indicator

Visual alert — flashes when gas concentration exceeds the threshold.

GSM / Wi-Fi Module

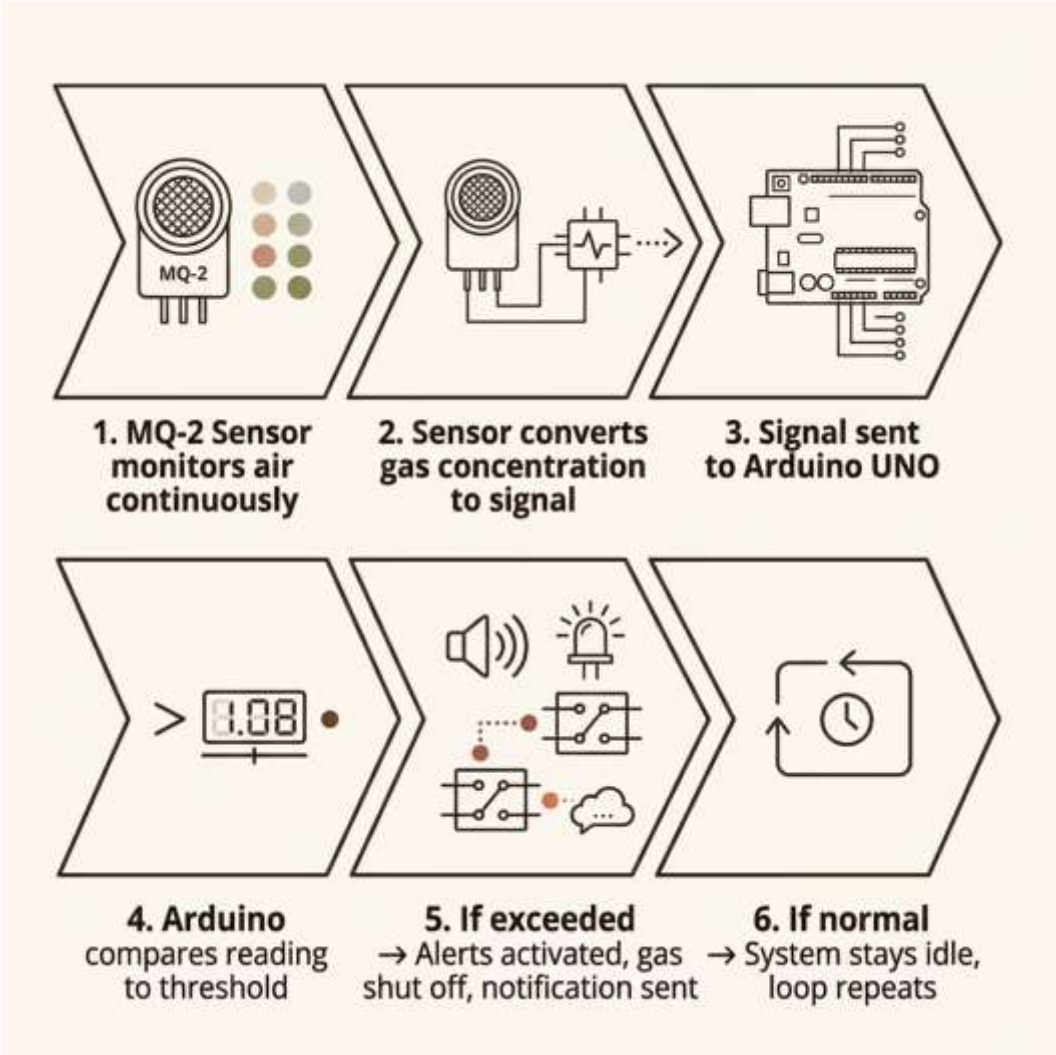
Optional — sends remote SMS or app notifications for off-site alerts.



LCD Display

Optional — shows real-time gas concentration readings for local monitoring.

Working Principle



How It Works

The MQ-2 sensor continuously samples the surrounding air. When LPG concentration rises, the sensor's output voltage increases proportionally.

The Arduino reads this analog signal and compares it against a calibrated threshold. If exceeded, it instantly activates all safety outputs — ensuring zero-delay response to a potential leak.

> The system operates in a continuous loop, providing 24/7 monitoring without manual intervention.

Algorithm & Flowchart

Step-by-Step Algorithm

01

Initialize System

Power on and configure all components.

02

Read Sensor

Acquire gas concentration value from MQ-2.

03

Check Threshold

Is gas value > safe limit?

04

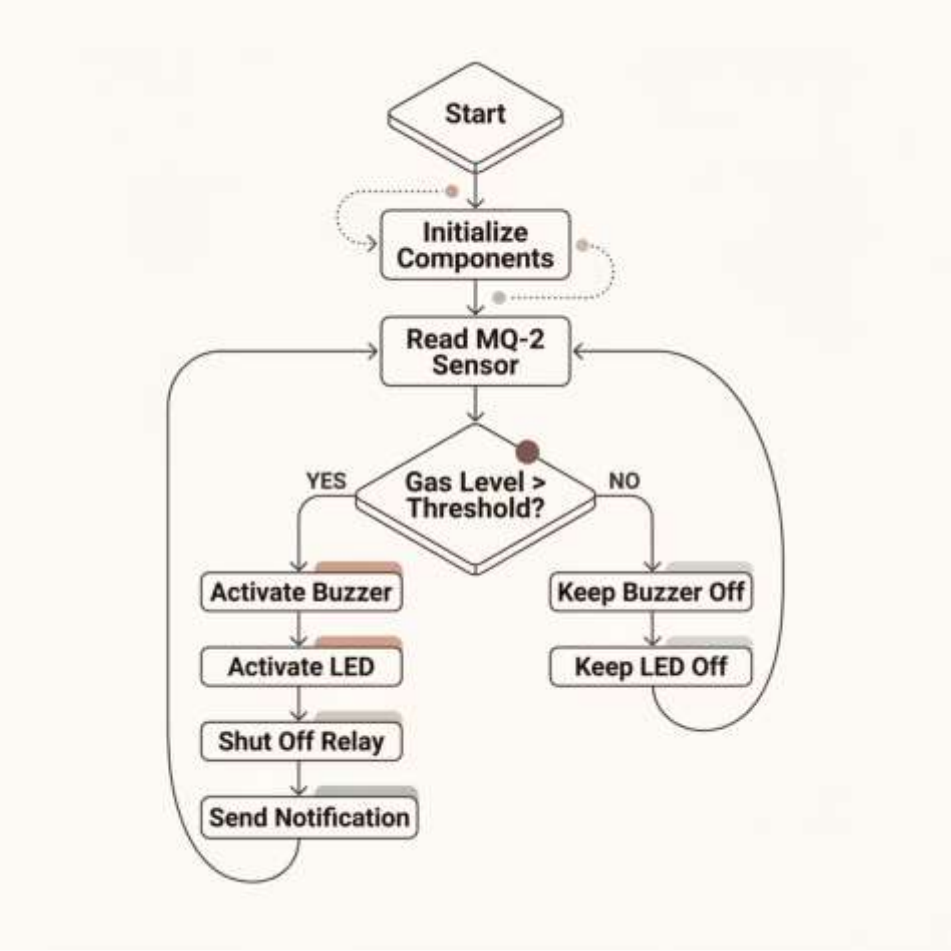
Alert & Act

Activate buzzer, LED, relay, and send notification.

05

Repeat

Return to Step 2 for continuous monitoring.



Applications

Domestic Use

Kitchens, homes, and apartments — preventing cooking-related gas accidents and protecting families.

Industrial Settings

Factories, laboratories, and LPG storage facilities requiring continuous, automated leak monitoring.

Commercial Establishments

Restaurants, hotels, and catering services where high-volume LPG use demands reliable safety systems.

Vehicle Safety

LPG-fueled vehicles benefit from onboard detection to prevent leaks in enclosed spaces.

Remote IoT Monitoring

Cloud-connected systems enable real-time alerts and remote safety management from anywhere.

References

[1] MQ-2 Gas Sensor Datasheet, Hanwei Electronics.

[2] Arduino UNO Reference Manual, Arduino LLC

[3] LPG Safety Standards, NFPA 58, National Fire Protection Association.

[4] IoT-Based Gas Monitoring Systems, IEEE Ac

Conclusion;

This system demonstrates that affordable, intelligent automation can significantly reduce LPG-related hazards — protecting lives, property, and infrastructure through real-time detection and immediate response.